Comprehensive Technical Analysis of the RoboSurvivor Autonomous Robotic Challenge 2026: Requirements, Comparative Competitions, Reference Systems, and Strategic Frameworks

The global landscape of autonomous robotics competitions has witnessed a paradigm shift toward challenges that integrate navigation, environmental perception, and complex object manipulation. The Autonomous Robotic Challenge (ARC) 2026, specifically the "RoboSurvivor" theme organized by the Universiti Malaya (UM) Makers Club, represents a high-water mark for educational mechatronics by requiring systems that are not only capable of traditional line-following but also of tactical decision-making and resource management within a non-linear field.[1] The requirements for this competition necessitate a robot with a multi-layered technical architecture, encompassing high-fidelity sensor arrays, robust mechanical manipulators, and advanced control algorithms like Proportional-Integral-Derivative (PID) loops to navigate varied path geometries.[3] This report analyzes similar global competitions, identifies mandatory robot attributes, synthesizes reference robot designs, and articulates a comprehensive winning strategy rooted in reliability and algorithmic adaptability.

Comparative Competition Ecosystems

To contextualize the RoboSurvivor 2026 challenge, one must examine existing autonomous robotics platforms that share technical DNA. These competitions serve as benchmarks for engineering standards and provide a library of proven mechanical and software solutions.

World Robot Olympiad (WRO) and the Future Engineers Category

The WRO is one of the most prominent parallels, particularly its "Future Engineers" and "RoboMission" categories. The WRO Future Engineers 2025/2026 season focuses on self-driving vehicles that navigate a parkours with randomized obstacles.[5] A key similarity lies in the autonomy requirement and the focus on the engineering process, though WRO often utilizes computer vision and Inertial Measurement Units (IMUs) rather than pure line following.[6] The "RoboMission" category aligns more closely with the ARC collection goals, requiring robots to push, lift, and grab game objects within a 2-minute run, compared to the 10-minute window afforded by the RoboSurvivor challenge.[1]

International Robot Olympiad (IRO): Robot Gathering

The "Robot Gathering" event at the IRO is perhaps the closest functional match to the ARC

resource collection phase. In this event, a robot must autonomously gather objects (typically cubes) in a designated area and deliver them to a final destination.[8] The mechatronics involved—infrared (IR) sensors for navigation and a mechanical arm for manipulation—mirror the fundamental requirements of the RoboSurvivor 2026 rulebook.[9] IRO competitors often face strict on-site construction rules, which emphasize the importance of modular mechatronic design.[9]

National Instruments Autonomous Robotics Competition (NI ARC)

Historically, the NI ARC has been a major influence on Malaysian robotics, with teams from Universiti Teknologi Malaysia (UTM) and Universiti Sains Malaysia (USM) frequently emerging as champions.[11] These competitions often utilized specific themes, such as "Mars Exploration" or "The Hospital of the Future," requiring robots to gather materials or deliver medicine accurately while avoiding obstacles.[12] Winning teams in this arena, such as Team Airost (UTM) and Team UOW, consistently prioritized reliability and stability over raw speed, a core insight applicable to the 2026 RoboSurvivor challenge.[11]

ABU Robocon and RoboCup Malaysia

ABU Robocon represents the elite tier of university robotics, involving highly creative solutions for tasks like ring tossing or complex assembly.[14] While ARC 2026 is less aggressive, the vision-based navigation and object tracking techniques developed for Robocon provide advanced reference points for color recognition and positional accuracy.[15] Similarly, RoboCup Malaysia Open serves as a local proving ground for autonomous systems, fostering the technical community that populates the ARC circuit.[2]

Comparative Table of Competition Constraints and Metrics

Competition	Primary Navigation	Primary Manipulation	Time Limit	Technical Complexity
ARC RoboSurvivor 2026	Compulsory Line Following [1]	Lifting/Dragging I-Shape [1]	10 Minutes [1]	High (Non-linear pathing)
WRO Future Engineers	CV / IMU / LIDAR [5]	Obstacle Avoidance [6]	3 Minutes [6]	Extreme (Full Autonomy)
IRO Robot Gathering	Line Following / Grid [10]	Gripper / Scoop [10]	Variable [8]	Medium (Speed focused)
NI ARC (Historical)	Multi-sensor localization [13]	Specific Block Placement [13]	Variable [12]	High (Hardware specific)

Mandatory Robot Technical Requirements for ARC 2026

The "RoboSurvivor" rulebook establishes rigorous technical boundaries. A robot that fails to meet these metrics during the registration and quarantine phases will be ineligible for competition rounds.[1]

Dimensional and Weight Restrictions

The initial sizing phase requires the robot to fit strictly within a box of $200mm \times 200mm \times 200mm$.[1] This constraint is critical because it forces engineers to consider folding or compact mechanisms that only deploy once the game commences. Crucially, the rulebook allows for expansion after the start signal, meaning a robot can extend its sensor outriggers or manipulator arms to encompass a larger footprint, provided it stays within the overall field boundaries.[1] The maximum weight is capped at 3.0 kg, which is relatively generous for a robot of this scale, allowing for robust metal gearboxes and high-capacity batteries.[1]

Power System Architecture

Safety is a primary concern in the electrical design. The robot must operate on a maximum of 12V, which typically translates to a 3S Lithium-Polymer (LiPo) battery.[1] Participants are prohibited from using damaged or swollen batteries, and charging is restricted to designated safety stations.[1] From a design perspective, this 12V ceiling provides ample power for high-torque servos and powerful drive motors, provided the wiring is rated for the expected current draw.[1]

Control and Autonomy Standards

The robot must be 100% autonomous. The rules strictly prohibit any form of external human control, including Bluetooth, Wi-Fi, IR, or RF signals meant for operation.[1] While single-board computers like the Raspberry Pi or Jetson Nano are permitted, standard microcontrollers such as Arduino (Uno, Mega, Pro Mini), ESP32, or STM32 are more common due to their real-time processing capabilities for sensor feedback.[1] Laptop-to-robot communication is strictly forbidden during the run, precluding real-time telemetry adjustments.[1]

Safety Mechanisms

A mandatory safety requirement is the inclusion of a red, latching Emergency Stop (E-Stop) button.[1] It must be positioned on the top of the robot, easily accessible to referees or operators. Activating this button must immediately cut all power to the motors to prevent damage to the field or injury to personnel.[1]

Prohibited Components and Restrictions

To maintain the integrity of the vinyl field and ensure a safe environment, certain materials and components are blacklisted:

- **Surface Protection:** Wheels and bases must not damage the PVC banner. Studded tires

and strong adhesives (e.g., duct tape) that leave residue are banned.[1]

- **Fluid and Pressure Limits:** Hydraulic systems and liquid-based cooling are prohibited to prevent leaks. Pneumatic systems are limited to a maximum pressure of 0.5 MPa.[1]
- **Signal Interference:** Devices designed to jam IR or Wi-Fi signals are strictly prohibited.[1]
- **Hazardous Materials:** Flammables, combustibles, and high-class lasers (Class 2 and above) are banned.[1]

Mechatronic System Design for Resource Management

The core of the RoboSurvivor challenge is the collection of "I-shaped" resources (Food) and their delivery to processing squares.[1] This requires a mechatronic design that balances speed, precision, and gripping reliability.

Drivetrain and Chassis Optimization

The mobility platform must navigate four distinct path types: straight lines, sine waves, sharp 90-degree zig-zags, and dashed lines.[1] A differential drive system—two independent drive wheels and a low-friction caster—is the standard for such tasks due to its ability to turn in place.[3]

Tire and Traction Considerations

The field is a printed PVC/Vinyl banner with specific friction characteristics.[1] Standard rubber tires may slip during the high-speed oscillations of the blue sine wave path. Some teams utilize silicone-coated wheels or specialized high-friction rubber to ensure consistent traction.[18] While suction fans for downforce are permitted, they must be carefully calibrated to ensure they do not "lift" or scratch the field surface, which would lead to disqualification.[1]

Center of Gravity and Structural Integrity

Maintaining a low center of gravity (CoG) is essential, especially when the robot is carrying a 50g resource at the end of an extended manipulator.[1] A rigid chassis construction minimizes mechanical flex, which can otherwise introduce noise into the sensor readings.[4] Topology optimization, a process of removing material from non-load-bearing areas, can help achieve a stiff yet lightweight base.[18]

Manipulator Engineering for I-Shaped Resources

The 3D-printed PLA resources feature a central 20mm cylinder capped by 30mm square blocks.[1] This unique geometry offers multiple gripping strategies.

Gripper Mechanics

An "encompassing grip" is significantly more reliable than a simple friction-based pinch.[20] Designers should aim for a gripper whose jaws match the 20mm diameter of the central

cylinder. By overlapping the square flanges, the robot achieves a positive mechanical lock, preventing the resource from rotating or slipping during transport.[20]

- **Three-Finger Adaptive Grippers:** These designs offer superior stability compared to two-finger models, particularly if the robot needs to pick up resources that have toppled over (which is permitted as long as they remain on the field).[1]
- **Soft and Woven Grippers:** Recent research suggests that woven plastic grippers can lift payloads many times their own weight while remaining lightweight and conforming to irregular shapes.[24] While perhaps overkill for a 50g resource, these designs represent the cutting edge of compliant mechatronics.
- **Material Selection:** Using Thermoplastic Polyurethane (TPU) pads on the gripper's inner surface provides a high coefficient of friction and a degree of compliance, safeguarding the PLA resource from being punctured or damaged by the actuator.[22]

Lifting and Delivery State

The rulebook defines a "Valid Grip" as lifting the resource completely off the ground or dragging it in a controlled manner.[1] Lifting is the preferred method, as it eliminates the friction between the resource and the banner, which could otherwise cause the robot to stall or derail on the yellow zig-zag path.[1] A single-DOF lifting arm powered by a high-torque servo (e.g., DS3225) is typically sufficient to elevate the resource a few centimeters for transport.[25]

Perception and Identification Systems

The robot must distinguish between five colors: black (Locks), red, green, blue, and yellow (Resources).[1]

Ground Color Detection

The field includes $60mm \times 60mm$ colored base indicators beneath each resource.[1] These markers are larger than the resources themselves to facilitate easier sensor identification. Advanced color sensors, such as the TCS3200 or the I2C-based TCS34725, are recommended for placement at the bottom of the chassis.[1] The TCS34725 is particularly valued for its superior performance under non-uniform indoor lighting.[29]

Line Following Sensor Arrays

A high-density IR array, such as the 8-channel Pololu QTR-8RC, is vital for navigating the 18mm black line.[3] These sensors work by measuring the reflection of IR light; white surfaces reflect more light (lower resistance/value), while the black vinyl line absorbs light (higher value).[28] Spacing the sensors such that they are closer together than the line width ensures there are no "blind spots" in the array.[4]

Proximity and Obstacle Detection

While the RoboSurvivor field is generally clear of obstacles other than the resources, proximity sensors on the gripper are useful for detecting when the robot is perfectly aligned with a

resource.[26] An ultrasonic sensor (HC-SR04) or a 2-meter Time-of-Flight (ToF) distance sensor can identify the "Lock" or "Food" from a distance, allowing the robot to slow down and prepare for a precise grip.[5]

Advanced Control and Navigation Algorithms

The diverse path profiles of ARC 2026—Straight, Dashed, Sine, and Zig-Zag—require more than a simple "bang-bang" controller. A sophisticated algorithmic approach is necessary to maintain speed without sacrificing accuracy.

Proportional-Integral-Derivative (PID) Control

PID control is the gold standard for high-speed line following. It transforms the erratic left-right corrections of basic controllers into smooth, graduated motor speed adjustments.[3]

Mathematical Basis of PID

The system calculates an error $e(t)$ based on the line's position relative to the center of the sensor array.[30]

Where s_i is the value of sensor i and w_i is its weight (e.g., 0 to 7000).[3] The PID output determines the speed differential between the left and right motors:

$$Output = K_p \cdot e(t) + K_i \int e(t)dt + K_d \frac{de(t)}{dt}$$

- **Proportional (P):** Directly responds to the current offset. If K_p is too high, the robot will oscillate; if too low, it will fail to turn in time for the sine wave.[4]
- **Integral (I):** Corrects steady-state error by accumulating past drift. This is useful for long curves but must be carefully limited to avoid "windup" and instability.[3]
- **Derivative (D):** Predicts future error by calculating the rate of change. This provides a "damping" effect that is crucial for maintaining stability on the blue sine wave path.[3]

Path-Specific PID Tuning

Competitive robots often use adaptive PID parameters for different track sections.[4]

- **Straight and Sine Paths:** High K_p and moderate K_d values allow for aggressive tracking of the smooth 100mm amplitude sine wave.[1]
- **Zig-Zag Path:** The 90-degree sharp turns of the yellow path cannot be handled by pure PID. Instead, the robot should detect the sharp angle using the outer sensors of the array and switch to a "turn-in-place" routine until the center sensors re-acquire the line.[36]

Bridging Gaps in the Dashed Line (Green Path)

The green path consists of 400mm segments separated by 50mm gaps.[1] A standard line follower might interpret a gap as a dead end or a derailment.

Gap-Verification Logic

To navigate gaps successfully, the robot must implement a verification routine.[33]

1. **State Detection:** When all sensors in the array return a "white" reading, the robot checks its previous state.[33]
2. **Logic:** If the robot was centered on the line immediately before the signal loss, it enters a "Gap Phase" rather than a "Derailment Phase".[33]
3. **Execution:** The robot continues to move straight forward at a fixed speed for a short duration (e.g., 200-300ms).[33]
4. **Re-acquisition:** If the line is detected again within the designated time limit, the robot resumes normal PID control. If not, it assumes it has reached a dead end and executes a U-turn or retry logic.[33]

Shortest Path and Node Logic

In a maze-like environment with multiple processing squares, the robot should treat junctions as "nodes".[38] By recording its direction at each node, the robot can optimize its return path from the processing squares to the junction, minimizing the time spent in transit.[38]

Game Flow and Strategic Frameworks

Winning the RoboSurvivor challenge requires a strategic understanding of the point values, randomized elements, and the mandatory "Unlock" mechanic.[1]

The "Unlock" Sequence and Logic

There are four black "Locks" blocking the entrance to each resource branch.[1] A critical rule is that a specific branch is only "Unlocked" once its black resource has been successfully delivered to the Black Processing Square at the top center of the field.[1] Points for colored resources are *only* awarded if the corresponding lock has been cleared.[1]

Tactical Sequencing

Robots can choose to unlock all branches first or proceed with an "unlock-and-collect" cycle for each branch.[1]

- **Efficiency:** Unlocking all branches first may reduce the number of times the robot has to travel to the top center "Black Zone." However, this increases the risk of carrying a lock while potentially knocking over colored resources in the junction.
- **The "Yellow" Priority:** Because the Yellow Zig-Zag path offers the highest delivery points (50 per resource), it is tactically sound to unlock and clear this branch first.[1] This ensures that even if the 10-minute timer expires, the team has secured the most difficult and

high-value points.

Scoring Rubric and Point Distribution

The scoring system is designed to reward both navigation and accuracy. A "Perfect Run" (all resources and locks delivered, robot returns to start) totals 495 points.[1]

Scoring Category	Specific Action	Points Awarded	Max Category Points
Navigation	Leave Start / Reach Junction	+5 / +10	15
Black Locks	Secure Grip / Deliver to Black Zone	+10 / +20	120 (4x30)
Resources	Secure Grip (any color)	+5	40 (8x5)
Red Delivery	Correct Square (Straight)	+20	40 (2x20)
Green Delivery	Correct Square (Dashed)	+30	60 (2x30)
Blue Delivery	Correct Square (Sine)	+40	80 (2x40)
Yellow Delivery	Correct Square (Zig-Zag)	+50	100 (2x50)
Bonus	Perfect Run Bonus	+40	40
Grand Total			**495**

Handling Randomization

The arrangement of colored resources is randomized once per round and applies to all teams.[1] However, the rulebook provides a "Randomization Logic" guarantee: the "First Row" (closest to the start) will contain exactly one of each color (1R, 1G, 1B, 1Y), and the "Second Row" will do the same.[1] This prevents a situation where one branch is overloaded, allowing teams to develop a robust "state machine" logic that searches for the next needed color without knowing the specific layout beforehand.[1]

The Retry Procedure: Risk vs. Reward

If a robot stalls for ≥ 20 seconds or derails, a "Retry" is mandatory.[1] The operator picks up the robot and returns it to the Start Zone.

- **Penalties:** Held colored resources are removed from play and cannot be scored.[1] However, Black Locks are returned to their initial position, necessitating a restart of the unlock process for that branch.[1]
- **Strategy:** Reliability is paramount. A slow, error-free run that avoids retries is often superior to a fast, erratic run that loses resources and forces multiple restarts.[13]

Comparative Analysis of Reference Robotic Systems

Studying the design choices of successful robots in similar autonomous challenges reveals common traits associated with victory.

Reference System 1: Team Apollo (WRO Future Engineers 2025)

Team Apollo's vehicle represents the high-tier integration of sensor fusion and real-time decision-making.[5]

- **Vision and Sensing:** It utilizes a Limelight 3A for color detection and a 9-axis IMU for drift correction.[5] This combination allows the robot to maintain a straight heading even when it loses visual landmarks, a strategy directly applicable to the dashed line gaps in ARC 2026.
- **Drive Mechanics:** The use of AndyMark NeveRest motors with planetary gearboxes provides high torque and integrated encoder feedback, ensuring precise wheel rotations for 90-degree turns.[5]
- **Software:** Their "Multifactor Decision System" blends IMU data, wall distances, and vision tracking to navigate complex environments, highlighting the importance of redundant sensor logic.[5]

Reference System 2: HBFS Robotics (Fastest Line Follower)

HBFS Robotics holds records for some of the world's fastest line followers, focusing on raw navigational performance.[19]

- **Chassis Design:** Their robots feature an extremely low profile and a very low center of gravity to prevent flipping during high-speed cornering.[40]
- **Sensor Arrays:** They often use 6 or 8-channel arrays positioned far in front of the drive wheels, providing a "look-ahead" distance that allows the PID controller more time to react to curves.[4]

Reference System 3: Rufus (NI ARC Winner)

The winner of the 2016 NI ARC focused on a "simple and reliable approach".[13]

- **Philosophy:** Rufus moved with "moderate speed," which allowed it to distribute objects successfully without bumping the field boundaries or dropping resources.[13]
- **Key Insight:** While other teams pushed for speed and crashed out, the UOW team waited for their competitors to make mistakes, securing victory through consistency.[13]

Reference System 4: BlindGrasp (Gripper Innovation)

BlindGrasp, a KUKA Innovation Award finalist, developed a novel gripper with tactile sensing capabilities.[41]

- **Manipulation:** By using sensors to detect contact and proximity, the gripper can adjust its force to the object's properties.[41] In the context of RoboSurvivor, such a system could ensure that the 3D-printed PLA resources are not crushed by excessive servo force.

Winning Strategy: A Multi-Phase Approach

To secure a victory in the ARC RoboSurvivor 2026, a team must execute a strategy that integrates mechanical reliability, algorithmic robustness, and tactical foresight.

Phase 1: Robust Environmental Calibration

Lighting and surface friction are the two most variable environmental factors.[1]

- **Dynamic Thresholding:** Rather than using a fixed value for "black" and "white," the robot should calculate the midpoint dynamically during its 10-second calibration phase.[3]
- **Voltage Compensation:** As the 12V LiPo battery drains over the 10-minute run, motor performance will degrade. Implementing a voltage regulator or monitoring battery levels to adjust PWM duty cycles ensures the PID parameters remain effective throughout the entire run.[4]

Phase 2: High-Stability PID Implementation

The goal is to achieve "critically damped" movement—reacting quickly to the line without overshooting.[4]

- **Tuning Sequence:** Start with P-only control until oscillation begins, then add Derivative (D) to dampen the motion.[4] The Integral (I) term should be used sparingly, only if the robot consistently rides to one side of the line due to mechanical imbalances.[4]
- **Adaptive Speed:** The robot should slow down when the error $e(t)$ increases, indicating a sharp curve or a zig-zag turn, and speed up on straight sections to maximize time efficiency.[4]

Phase 3: Tactical Path Planning

The robot's "brain" should be a robust Finite State Machine (FSM).

- **Task Prioritization:** Unlock the high-scoring branches (Yellow and Blue) first. This strategy front-loads the score, providing a buffer if mechanical issues arise later in the run.[1]
- **The "Cargo" Advantage:** If the mechanical design permits, carrying two resources at once can drastically reduce the number of laps required to clear the field.[1] A "cargo bed" or a multi-stage manipulator that can hold one resource while picking another is a significant advantage.
- **Retry Recovery:** The FSM must use the robot's internal memory (EEPROM) to store its progress. If a retry is called, the robot should be able to resume its task list from the current state (e.g., knowing which branches are already unlocked) rather than restarting the entire loop from zero.[1]

Phase 4: Finalist Pitching and Communication

For the Top 5 finalists, the pitching session is worth 50 points—a significant portion of the total score.[1]

- **Content Guidelines:** Teams must clearly explain their engineering rationale, software strategy (e.g., their PID tuning method), cost analysis, and innovation.[1]

- **Presentation Best Practices:** Use a "one-page technical summary" with high-quality photos of the robot and diagrams of the code logic.[45] Judges prioritize teams that demonstrate a deep understanding of their system, often referred to as "Anti-Ghost Building".[1]
- **Transparency regarding Challenges:** Be open about the obstacles the team faced and how they were solved. Judges value resilience and the engineering journey as much as the final score.[47]

Synthesis of Operational Mechanics and Strategic Outcomes

The success of the "RoboSurvivor" depends on the delicate balance between mechatronic capability and algorithmic intelligence. The compulsory line-following rule precludes the use of advanced SLAM (Simultaneous Localization and Mapping) or coordinate-based navigation, forcing a reliance on the 18mm black line as the primary source of truth.[1] Consequently, the most successful robots will be those that can "feel" the line with the greatest sensitivity and manipulate the I-shaped resources with the highest degree of mechanical security.

The transition from a "Start Zone" to the "Black Processing Zone" to unlock a branch is a mandatory gatekeeper that ensures only robots with both navigation and manipulation capabilities can score.[1] The scoring rubric heavily favors those who can complete the course quickly, with tie-breakers decided by run time and resource count.[1] Ultimately, the winner will likely be a team that minimizes "toppled" resources, avoids "derailment" through robust PID damping, and maximizes the 10-minute window through efficient multi-resource transport and intelligent state management.

By integrating the lessons from WRO, IRO, and historical NI ARC events, a RoboSurvivor team can build a system that is prepared for the non-uniformities of a real-world field. Consistent practice, dynamic sensor calibration, and a reliability-first philosophy are the foundations of a championship performance in the Universiti Malaya makerspace.[1]

Works cited

1. PARTICIPANT_RULEBOOK_ARC_RoboSurvivor_2026.pdf
2. REC, accessed February 26, 2026, https://rec-um.github.io/
3. Line Follower Robot (with PID controller) - Arduino Project Hub, accessed February 26, 2026, https://projecthub.arduino.cc/anova9347/line-follower-robot-with-pid-controller-01813f
4. PID Tuning for Line Follower Robot: Complete How-To Guide - ThinkRobotics.com, accessed February 26, 2026, https://thinkrobotics.com/blogs/learn/pid-tuning-for-line-follower-robot-complete-how-to-guide
5. oracleapolloo/WRO-FUTURE-ENGINEERS-2025 - GitHub, accessed February 26, 2026, https://github.com/oracleapolloo/WRO-FUTURE-ENGINEERS-2025

6. WRO Future Engineers Category – Game Rules 2026, accessed February 26, 2026, https://wro-association.org/wp-content/uploads/WRO-2026-Future-Engineers-Self-Driving-Cars-General-Rules.pdf
7. 2025 Season - WRO Association, accessed February 26, 2026, https://wro-association.org/competition/2025-season/
8. Robot Gathering, accessed February 26, 2026, https://iroc.gr/pdf/Robot-Gathering.pdf
9. Game description Robot Gathering aims to arrive at the point of destination to arrive quickly collects the entire object given - IISRO, accessed February 26, 2026, https://iisro.co/wp-content/uploads/2023/08/IISRO-2023_Robot-Gathering.pdf
10. IRO International Robot Olympiad_Robot Gathering, accessed February 26, 2026, https://www.youtube.com/watch?v=32AbmFC_VeU
11. UTM Students Champion for 2 Years at NI Autonomous Robotics Competition (ARC) 2022, accessed February 26, 2026, https://news.utm.my/2022/12/utm-students-declared-as-champion-again-in-ni-autonomous-robotics-competition-arc-2022/
12. National Instruments Accelerates STEM Education Through A Robotics Competition, accessed February 26, 2026, https://amcham.com.my/national-instruments-accelerates-stem-education-through-a-robotics-competition/
13. Student team crowned champions in autonomous robotics competition - University of Wollongong, accessed February 26, 2026, https://www.uow.edu.au/media/2016/student-team-crowned-champions-in-autonomous-robotics-competition.php
14. Robocon - REC, accessed February 26, 2026, https://rec-um.github.io/Robocon.html
15. (PDF) A Robot Vision System in ABU Robocon Contest: Visual Servoing Navigation and Object Tracking - ResearchGate, accessed February 26, 2026, https://www.researchgate.net/publication/272054773_A_Robot_Vision_System_in_ABU_Robocon_Contest_Visual_Servoing_Navigation_and_Object_Tracking
16. ROBOCUP MALAYSIA OPEN 2024 - YouTube, accessed February 26, 2026, https://www.youtube.com/watch?v=Udwoip8a4IM
17. WRO Future Engineers Getting Started, accessed February 26, 2026, https://world-robot-olympiad-association.github.io/future-engineers-gs/
18. I Need Advice to Make a Fast Line Following Robot For a Competiton - Reddit, accessed February 26, 2026, https://www.reddit.com/r/AskRobotics/comments/1p5k6mq/i_need_advice_to_make_a_fast_line_following_robot/
19. HBFS Line Follower at Robotex 2017 - YouTube, accessed February 26, 2026, https://www.youtube.com/watch?v=d8eesFMPTeE
20. Introduction To Designing Robot Gripper Finger Tooling - GripShape, accessed February 26, 2026, https://www.gripshape.com/resources/introduction-to-designing-robot-gripper-finger-tooling

21. Robotic Arm Gripper for FRC Logomotion Competition : 4 Steps - Instructables, accessed February 26, 2026, https://www.instructables.com/Robotic-Arm-Gripper-for-FRC-Logomotion-Competition/
22. An Open-Source 3D Printed Three-Fingered Robotic Gripper for Adaptable and Effective Grasping - MDPI, accessed February 26, 2026, https://www.mdpi.com/2313-7673/10/1/26
23. A novel soft gripper with enhanced gripping adaptability based on spring-reinforced soft pneumatic actuators | Industrial Robot | Emerald Publishing, accessed February 26, 2026, https://www.emerald.com/ir/article/50/4/595/183540/A-novel-soft-gripper-with-enhanced-gripping
24. This Low-Cost Woven-Plastic Robotic Gripper Can Lift Nearly 800 Times Its Own Weight, accessed February 26, 2026, https://www.hackster.io/news/this-low-cost-woven-plastic-robotic-gripper-can-lift-nearly-800-times-its-own-weight-91854883ca67
25. 3D Printable Modular Robot Gripper - Indystry.cc, accessed February 26, 2026, https://indystry.cc/product/3d-printable-modular-robot-gripper/
26. Line-Following Robot With Obstacle Avoidance : 9 Steps - Instructables, accessed February 26, 2026, https://www.instructables.com/Line-Following-Robot-With-Obstacle-Avoidance/
27. design and fabrication of an autonomous line follower robot capable of picking and dropping objects from - Kenyatta University, accessed February 26, 2026, https://ir-library.ku.ac.ke/server/api/core/bitstreams/f7536db1-4d0f-40fa-b120-95e5cca96e34/content
28. Sorting System Based on Color Using Line Follower Robot - Semantic Scholar, accessed February 26, 2026, https://pdfs.semanticscholar.org/f2e3/4d9eaba0c81ce7dda3eaea98a5c4b3801902.pdf
29. COLOR SORTING AND LINE FOLLOWING ROBOT - iarjset, accessed February 26, 2026, https://iarjset.com/wp-content/uploads/2024/12/IARJSET.2024.111221.pdf
30. PID Tutorials for Line Following - RobotShop Community, accessed February 26, 2026, https://community.robotshop.com/forum/t/pid-tutorials-for-line-following/13164
31. Line Follower Robot Using PID Algorithm | Full Electronics Project, accessed February 26, 2026, https://www.electronicsforu.com/electronics-projects/hardware-diy/line-follower-robot-using-pid-algorithm
32. Robot - Line Follower : 35 Steps (with Pictures) - Instructables, accessed February 26, 2026, https://www.instructables.com/Robot-Line-Follower/
33. Program for line following robot with dotted line - Electronics - Pololu ..., accessed February 26, 2026, https://forum.pololu.com/t/program-for-line-following-robot-with-dotted-line/24816
34. PID Line Follower Tuning - Robot Research Lab, accessed February 26, 2026,

https://robotresearchlab.com/2019/02/16/pid-line-follower-tuning/
35. Tuning PID controller for sharp turns in line follower robot - Robotics Stack Exchange, accessed February 26, 2026, https://robotics.stackexchange.com/questions/11292/tuning-pid-controller-for-sharp-turns-in-line-follower-robot
36. Algorithm for Line Follower Robots to Follow Critical Paths with Minimum Number of Sensors - CMU, accessed February 26, 2026, https://www.andrew.cmu.edu/user/yuchingw/229655734.pdf
37. Gapped & Crossed Line Following. Part 4. Follow to Gap Pro Preview - FLLCasts, accessed February 26, 2026, https://www.fllcasts.com/tutorials/207-gapped-crossed-line-following-part-4-follow-to-gap
38. Shortest Path Line Follower Robot Logic Revealed! - EmbedJournal, accessed February 26, 2026, https://embedjournal.com/shortest-path-line-follower-robot-logic-revealed/
39. Line Follower Robot - King of Maze : 9 Steps - Instructables, accessed February 26, 2026, https://www.instructables.com/Line-Follower-Robot-King-of-Maze/
40. Line Follower Robot Race Competition 2024 | Fast LFR Competition in Bangladesh, accessed February 26, 2026, https://www.youtube.com/watch?v=S05KutbNJ4A
41. New robot gripper lets robots "see" with touch using neural networks - YouTube, accessed February 26, 2026, https://www.youtube.com/watch?v=pmWLGsZlmDk
42. Advanced Simulation Enables Better Robot-Object Interactions - Tech Briefs, accessed February 26, 2026, https://www.techbriefs.com/component/content/article/53536-advanced-simulation-enables-better-robot-object-interactions
43. How to ensure a 90 degree turn in a line follower robot - Quora, accessed February 26, 2026, https://www.quora.com/How-do-I-ensure-a-90-degree-turn-in-a-line-follower-robot
44. PID Tutorials for Line Following | RobotShop Community, accessed February 26, 2026, https://community.robotshop.com/blog/show/pid-tutorials-for-line-following
45. Best Practices for FIRST Robotics Competition Judged Awards, accessed February 26, 2026, https://community.firstinspires.org/hubfs/web/program/frc/awards/best-practices-for-teams.pdf
46. WRO2025 RoboMission Highlights | PDF | Robot - Scribd, accessed February 26, 2026, https://www.scribd.com/document/994686058/WRO2025-RoboMission-Highlights
47. Tournament Tips: Preparing for Your Judging Session - FIRST Community Blog, accessed February 26, 2026, https://community.firstinspires.org/tournament-tips-preparing-for-your-judging-session